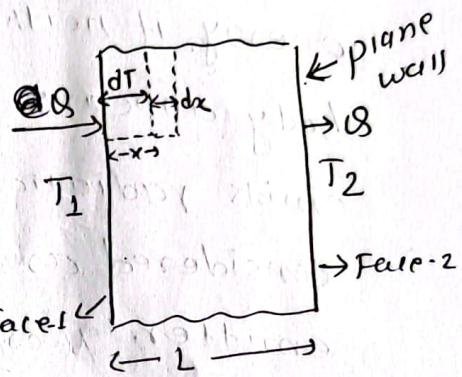
It is expressed as $-\frac{dT}{dx}$ and its unit is kelvin per meter (K/m). Here dT is the small change in temp. over a small distance dx and -ve sign indicates that temp. falls with distance.

One dim'l steady state heat conduction through plane wall

Consider parallel sided plane wall of thickness L and uniform cross-section area A. as shown in figure. Let K be the thermal conductivity of wall material through which heat is flowing only in x-direction.



[Fig: plane wall]

Let T_1 and T_2 are the temp. of higher temp. face-1 and lower temp. face-2. The small change of temp. dT when the particles conduct ~~the~~ every small distance dx.

According to fourier law, the rate of heat transfer is given by

$$\frac{dQ}{dt} = -KA\frac{dT}{dx}$$

$$\text{or } \frac{dQ}{dt}dx = -KA\,dT \quad \text{—(1)}$$

To obtain the rate of flow of heat through this wall is obtained by integrating eqn (1), we get

$$\int_0^L \frac{dQ}{dt}dx = \int_{T_1}^{T_2} -KA\,dT$$

$$\text{or } \frac{dQ}{dt}\int_0^L dx = -KA\int_{T_1}^{T_2} dT$$

$$\text{or } \frac{dQ}{dt}[L-0] = -KA[T_2-T_1]$$

$$\text{or } dQ/dt = \frac{KA(T_1-T_2)}{L}$$

$$\Rightarrow \frac{dQ}{dt} = \frac{(T_1-T_2)}{L/KA} \quad \text{—(2)}$$

where $\frac{L}{KA}$ = Thermal resistance R_{th}

$$\therefore \frac{dQ}{dt} = \frac{T_1-T_2}{R_{th}} \quad \text{—(3)}$$

This eqn (3) gives the rate of flow of heat through the wall. If R_{th} increases, $\frac{dQ}{dt}$ decreases and vice-versa.

Black body and black body radiation

A perfect black body is one which absorbs heat radiations of all wavelengths falling on it and emits radiations of all possible wavelengths when heated.

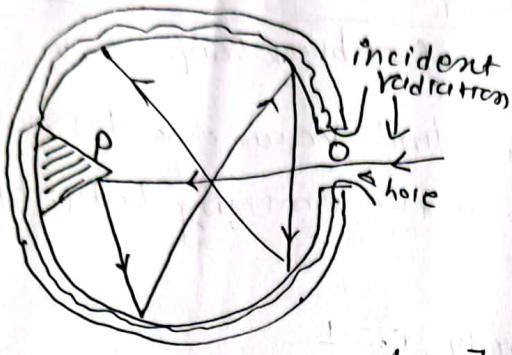
For a perfect black body absorptance (a) = 1 and emissivity (e) = 1. When a body absorbs all radiant energy, it neither reflect nor it transmit. As a result body appears black and known as black body. Sun emits radiations of all wavelengths. So it can be considered as a black body. A good absorber is a good emitter. Therefore, a black body is a good emitter and can emit the radiation of all possible wavelengths.

A perfect black body cannot be realized in actual practice. The nearest approach to a perfectly black body is a surface coated with lamp black or platinum black. Such a surface absorbs 96% to 98% of the incident radiation. Ferry's body is an example of perfectly black body realized in practice.

Ferry's black body:

Ferry's black body is a close approximation of perfectly black body which can be realized in practice. Ferry's body consists of a double walled hollow copper sphere with a small fine hole O and a conical projection P opposite to O as shown in figure.

A heating filament is kept in between two ~~walls~~ walls of the sphere and inner surface of sphere is coated with lamp black or platinum black.



[Fig: Ferry's black body]

When heat radiation falls on the conical projection P through hole O, projection P prevents the direct reflection of incident radiation rather it causes multiple reflections of radiations inside the sphere. Most of the incident radiation is absorbed by lamp black or platinum black and remaining part is also lost by multiple reflection. When the sphere is heated by passing electric current to the heating filament, radiations of all possible wavelengths come out from the hole O. Hence, the hole O behaves as a black body.

Emissive power

It is the amount of heat radiations radiated by a unit area of a black body per unit time at a particular temp. It is denoted by E. Mathematically,

$$\text{Emissive power } E = \frac{\text{Energy radiated } (Q)}{\text{Area } (A) \times \text{time } (t)} = \frac{Q}{A \times t}$$

$$\therefore E = \frac{Q/t}{A} = \frac{P}{A}$$

$$\text{unit of } E = \frac{J/sec}{m^2} = W/m^2.$$

According to Stefan's-Boltzmann law,

$$E = \sigma T^4 \quad \text{where } \sigma = \text{Stefan-Boltzmann constant} = 5.67 \times 10^{-8}\ W/m^2 K^4.$$

Emissivity (e):

Emissivity of a body is defined as the ratio of emissive power of a given body to the emissive power of a perfectly black body of same size at the same temp. It is denoted by e.

$\therefore e = \dfrac{\text{emissive power of body}}{\text{emissive power of perfect black body.}} = \dfrac{E}{E_b}$ or $\dfrac{P}{A \cdot t} = e$

Emissivity has no unit and dim^n. Value of e depends on the nature of the surface of the emitting body. It ranges from 0 to 1.

For a perfectly black body, $e = 1$

" a black body $e < 1$

For a perfectly reflecting surface $e = 0$.

$\therefore P =$

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Stefan's-Boltzmann's law of black body radiation:

Stefan's declared experimentally that energy radiated by a body depends only on its temp. and he put forward a law known as Stefan's law.

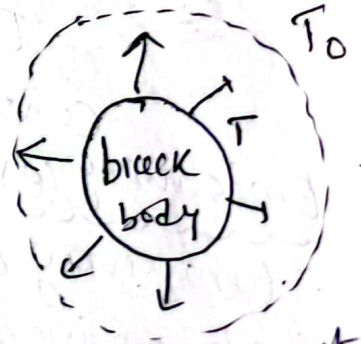
Stefan's-Boltzmann law states that " emissive power of a black body is directly proportional to the fourth power of absolute temp. of surface of the body".

let E be the emissive power of a black body at absolute temp. T. Then, according to Stefan's-Boltzmann law $E \propto T^4$

or $E = \sigma T^4$ — (1) where σ is proportionality constant is called Stefan's Const. The value of σ in SI system is $5.67 \times 10^{-8}\ Wm^{-2}K^{-4}$.

If a body is not perfectly black, and has its emissivity e. Then,

$E = e\sigma T^4$. — (2)



[Fig: Stefan's law of black body radiation

or $\frac{Q}{A \cdot t} = e\sigma T^4$ or $\frac{Q/t}{A} = e\sigma T^4 \Rightarrow \frac{\text{power}}{A} = e\sigma T^4$

$\therefore P = eA\sigma T^4$ — (3)

This relations given above are valid only if the temp. of the surrounding is zero kelvin. But, if there is a surrounding at temp T_0 (more than 0K.) then, surrounding also emits heat radiation, which is absorbed by black body. Hence, relation is modified by Boltzmann for the net emissive power as

$E' = \sigma(T^4 - T_0^4)$ — (4) This is called Stefan's-Boltzmann law.

If a body is not perfect black body, the net emissive power $E' = e\sigma(T^4 - T_0^4)$

$$\frac{Q'}{A \times t} = e\sigma(T^4 - T_0^4)$$

$$\text{or } \frac{Q'/t}{A} = e\sigma(T^4 - T_0^4)$$

$\therefore P' = eA\sigma(T^4 - T_0^4)$ — (5)

This gives power of radiated energy of a black body.

<u>Some Numerical problems</u>

(1) An insulating material having a thermal conductive of 0.08 W/m-K is used to limit the heat transfer of 80 W/m2 for a temp. of 15°C across the opposite faces. Find the thickness of the material.

Sol^n: $K = 0.08$ W/m-K, $\frac{Q}{tA} = 80$ W/m^2 $\Delta T = 15°C = 15K$

$dx = ?$ $\Rightarrow \frac{Q}{t} = KA\frac{dT}{dx} \Rightarrow dx = \frac{Q}{tAKdT} \Rightarrow dx = 0.15m$

(2) A brick wall 12 cm thick and $5m^2$ surface area exposed to 50°C at one face and 20°C to another face. If the thermal conductivity of the material is 1.5 W/m-K. Find the heat transfer rate.

Sol^n: $\Delta x = 12cm$, $A = 5m^2$ $dT = 50°C - 20°C = 30°C = 30K$.

$k = 1.5 W/m\text{-}K$, $\frac{Q}{t} = ?$ $\frac{Q}{t} = KA\frac{dT}{dx} = 1875$ watt.

(3) Find the rate of heat lost from a brick wall ($k = 0.7 W/m$) of length 5m, height 4m and 0.25 m thick. The temp. of the inner surface 50°C and that of outer surface is 30°C. Also, calculate the distance from the inner surface at which the temp. is 40°C

Sol^n: k of brick wall = 0.7 W/m-K.

Area $A = 5m \times 4m = 20m^2$,

$dx = 0.25m$, $dT = 50°C - 30°C = 20°C = 20K$

$\frac{dQ}{dt} = KA\frac{dT}{dx}$ $\boxed{= 1120 \text{ watt}}$

Again, $dx' = ?$ where $\frac{dQ}{dt} = \frac{KA(T_1 - T_2)}{dx'} \Rightarrow 1120 = \frac{0.7 \times 20(50-4}{dx'}$

(4) A 1.2m long tube with outer diameter of 4cm outside temp of 120°C is exposed to an ambient air at 20°C. If the heat transfer coefficient between the tube surface and the air is 20 W/m²K. Find the rate of heat transfer from the tube to the air.

Sol^n: $dx = 1.2m$

$dT = 120°C - 20°C = 100°C = 100K$, Heat transfer rate $h = \frac{K}{dx} = 20 W/m^2$

$A = 2\pi R dx = 2\pi \times 2 \times 10^{-2} \times 12 = 0.1508 m^2$.

$\frac{dQ}{dt} = \frac{K}{dx} dT \cdot A = 301.593$ watt

(5) A furnace inside temp of 2250K has a glass circular viewing of 6cm diameter. If the transmissivity of glass is 0.08 makes calculations for the heat loss from the glass window due to radiation.

Sol^n: $A = \frac{\pi}{4}d^2 = \frac{\pi}{4}(6\times10^{-2})^2$, $E = \sigma T^4 \Rightarrow \frac{Q}{t \cdot A} = \sigma T^4 \times e$ per unit per

$e = 0.08$ $T = 2250K$ $\Rightarrow \frac{Q}{t} = \sigma T^4 e \cdot A = 328.59$ watt.